

Technology Stack

Date	2 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy
Maximum Marks	

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

The table below details the specific technologies chosen for each component of the FitBuddy architecture, with a brief justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS and Jinja2 templates	Jinja2 renders dynamic pages directly from FastAPI, connecting the frontend with the backend without a separate JavaScript framework. This keeps the interface simple, lightweight and user-friendly.
2	Backend / Server-Side	Python with FastAPI (served using Uvicorn)	FastAPI offers fast processing, clean routing and easy handling of user input. Python also integrates smoothly with the Gemini AI library and suits the team's skills.
3	AI / GenAI Layer	Google Gemini AI (Gemini API via Python SDK)	Generates personalised fitness responses and recommendations from user input. The Gemini model was researched and selected for its suitability to the project.
4	Database / Data Storage	No external database (user input handled at runtime)	FitBuddy processes user inputs on request and passes them to Gemini, so a database is not required for the core functionality. This keeps the application simple and easy to run locally.
5	Cloud / Hosting / Deployment	Local deployment (Uvicorn on localhost)	The application is prepared and tested for local deployment, which is simple to set up, needs no hosting cost and is sufficient for demonstration and testing.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Version Control & Development Tools	Git / GitHub, VS Code, Python virtual environment	Enables seamless team collaboration and code sharing, while the virtual environment keeps dependencies isolated and the development environment consistent across members.
7	Third-Party APIs / Other Tools	Google Gemini API (API key)	Enables the AI-powered features of FitBuddy, such as generating personalised workout and fitness guidance for the user.